

WASP-12b

R-1115

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_200303 · created 2026-07-30T21:11:20+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458911.76415 +/- 0.00071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.029 +/- 0.045
Transit depth (Rp/Rs)^2	0.084 +/- 0.26 [%]
Orbital Inclination (inc)	80.35 +/- 2.88
Airmass coefficient 1 (a1)	1.87 +/- 1.15
Airmass coefficient 2 (a2)	-0.254 +/- 0.088
Scatter in the residuals of the lightcurve fit is	69.33 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	6.75
Transit Duration (day)	0.103 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.029 ± 0.045 vs 0.1178 (deviation 0.97σ)
Duration fit vs published	0.103 d vs 0.125 d (deviation 0.98σ)
Mid-time O–C	0.8 min
Depth SNR	0.3
Residual scatter	69.33 %

Light curve

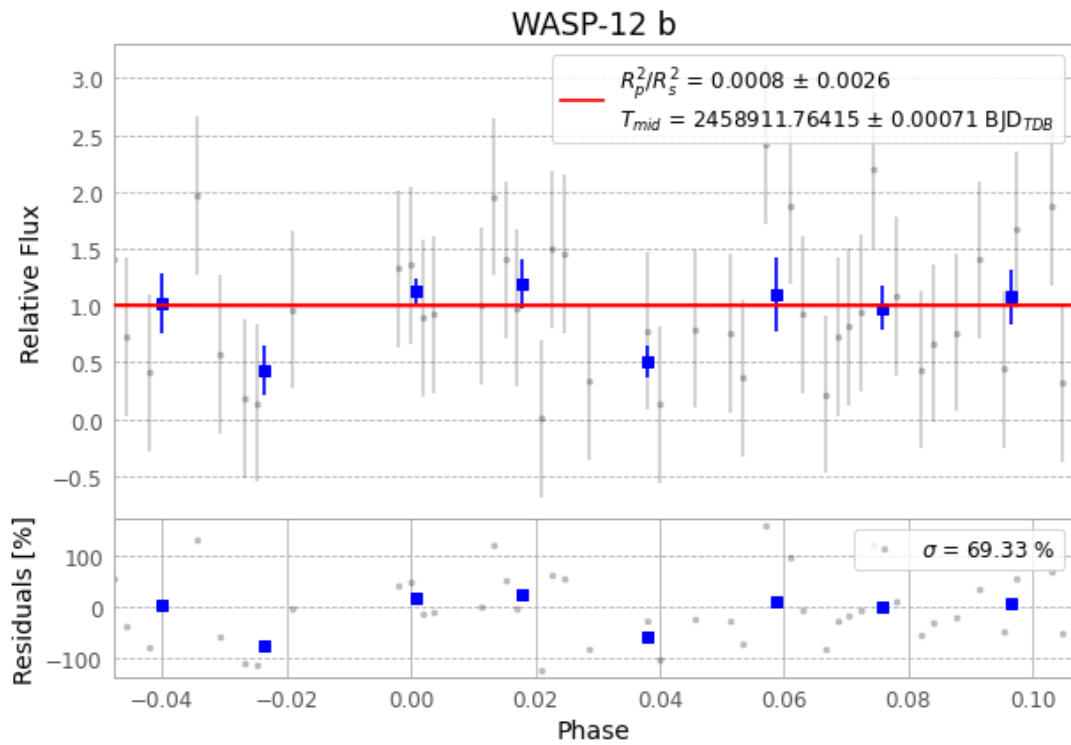


Figure 1 — FinalLightCurve_WASP-12 b_2020-03-02.png (SHA-256 2c8d743ee8a9bdae...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [542, 279], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 08c7d1029f6751a0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

90 FITS frames (59 MB), WASP-12200303043612.FITS → WASP-12200303090312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-200303.log (dbef1764c4f1...), FinalLightCurve_WASP-12 b_2020-03-02.png (2c8d743ee8a9...), FinalLightCurve_WASP-12 b_2020-03-02.csv (c66ab491a6a8...)

Compute resources

Wall time 180.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 21:11Z.